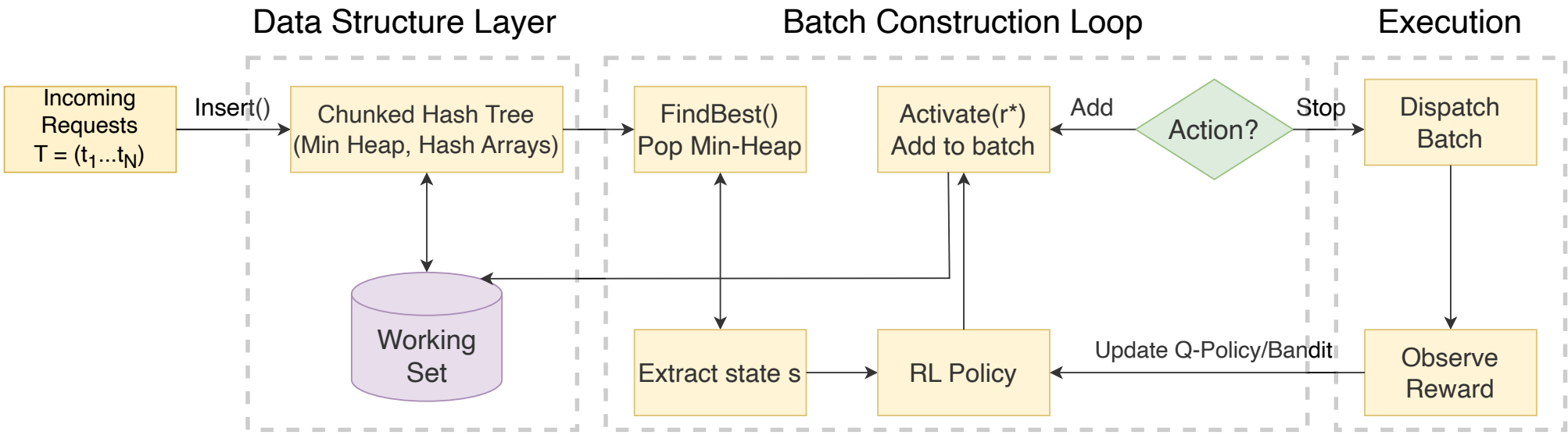

Data Structure Layer
Batch Construction Loop
Execution
Incoming Requests
T = (t1...tN)
Insert()
Chunked Hash Tree
(Min Heap, Hash Arrays)
Working Set
FindBest()
Pop Min-Heap
Extract state s
Activate(r*)
Add to batch
RL Policy
Add
Action?
Stop
Update Q-Policy/Bandit
Dispatch Batch
Observe Reward